

Gene-based Pan-genome Analysis of Cucumber

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Abstract

Global challenges such as climate change and population growth have intensified the need for more robust crop varieties. Advances in DNA sequencing technologies have made it easier to generate pangenomes from representative individuals of target species. Pangenomic studies have also helped identifying beneficial traits and provided insight into breeding efforts to develop more robust crop varieties (Schreiber et al., 2024). These studies have demonstrated success in recent plant research, uncovering evolution and diversity of different plant species (Tang et al., 2022) and identifying deleterious mutations to facilitate breeding (Wu et al., 2023). Cucumber (*Cucumis sativus* L.) is not only one of the most important vegetable crops worldwide, but also serves as a model plant for sex determination and vascular biology studies (Li et al., 2019). Previous studies have shown patterns in cucumber domestication, such as reduction in genetic diversity (Qi et al., 2013). These studies also highlight key genes associated with important agronomic traits and the history of the domestication process, providing valuable information for future breeding programs (Li et al., 2022). To capture the entire genomic diversity for future research and breeding, a complete gene-based pangenome from diverse cucumber accessions is needed.

Background

- Cucumber is one of the most important agricultural crops worldwide
- Different cucumber species/accessions have unique genes to express unique traits for climate adaption, and great potential for future breeding
- Pangenomic studies have advantages over single reference genome approaches, as a pangenome covers the complete genetic variation of the target species.

Materials

- Assembled genomes of 39 cucumber accessions covering different breeding status: Wild (10), Landrace (8), Cultivar (21), and from 10 different countries (Figures 1 and 2).

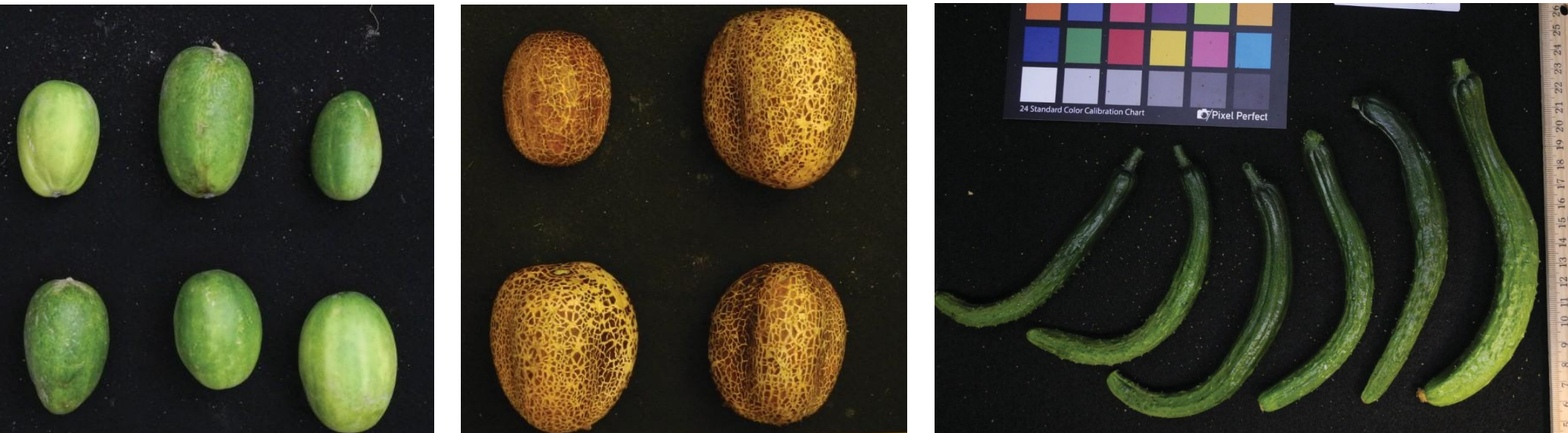


Figure 1(a) Wild Figure 1(b) Landrace Figure 1(c) Cultivar

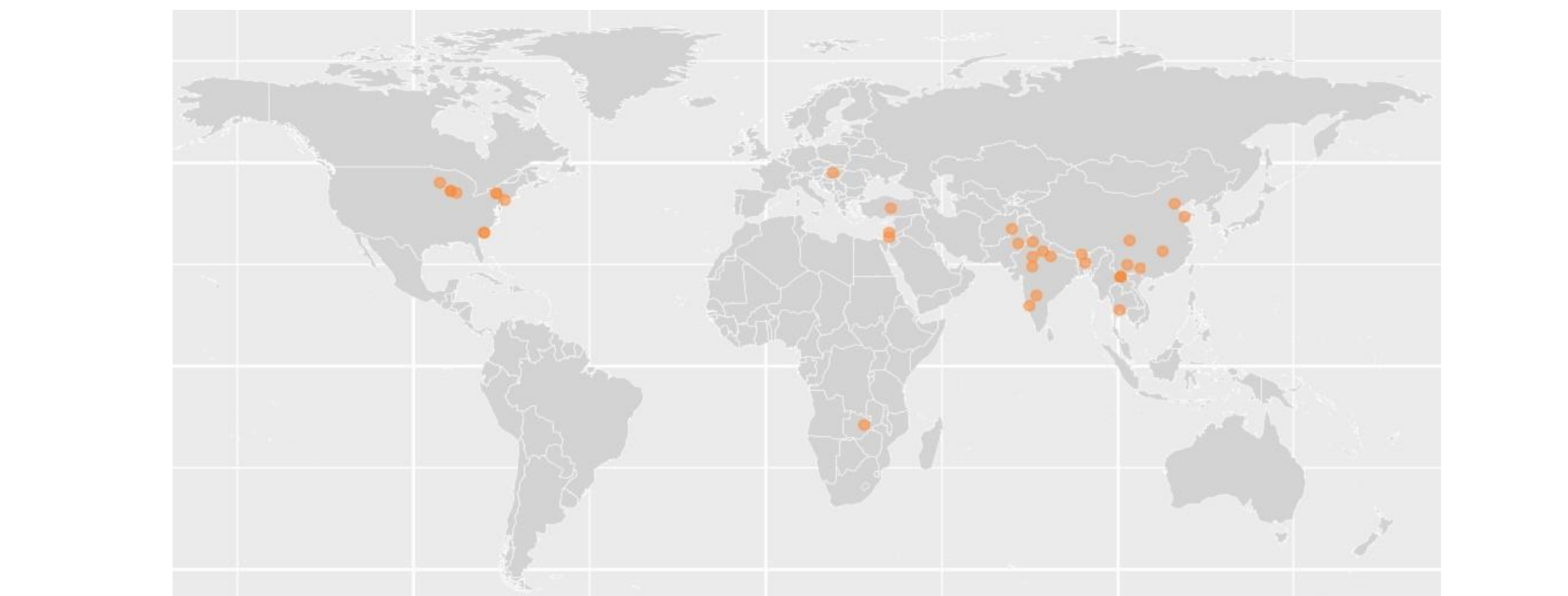


Figure 2: Originating locations of the sampled cucumber accessions

Summary

- Constructed a gene-based pangenome from 39 cucumber genomes.
- Identified 15,924 core genes, 2,474 soft core genes, 5,235 shell genes and 112 cloud genes in the cucumber gene-based pangenome.
- Identified 3,451 genes with significantly different presence frequencies between wild and landraces, enriched with those involved in energy production and cellular organization; and 2,895 genes between landrace vs. cultivars, enriched with those involved in plant growth and stress responses.

References

- Li, H., et al. (2022). Graph-based pan-genome reveals structural and sequence variations related to agronomic traits and domestication in cucumber. *Nature Communications*. 13, 682
- Li, N., et al. (2023). Super-pangenome analyses highlight genomic diversity and structural variation across wild and cultivated tomato species. *Nature Genetics*. 55, 852-860
- Qi, J., et al. (2013). A genomic variation map provides insights into the genetic basis of cucumber domestication and diversity. *Nature Genetics*. 45, 1510-1515
- Schreiber, M., et al. (2024). Plant pangenomes for crop improvement, biodiversity and evolution. *Nature Reviews Genetics*.
- Tang, D., et al. (2022). Genome evolution and diversity of wild and cultivated potatoes. *Nature*. 606, 535-541

Workflow & Results

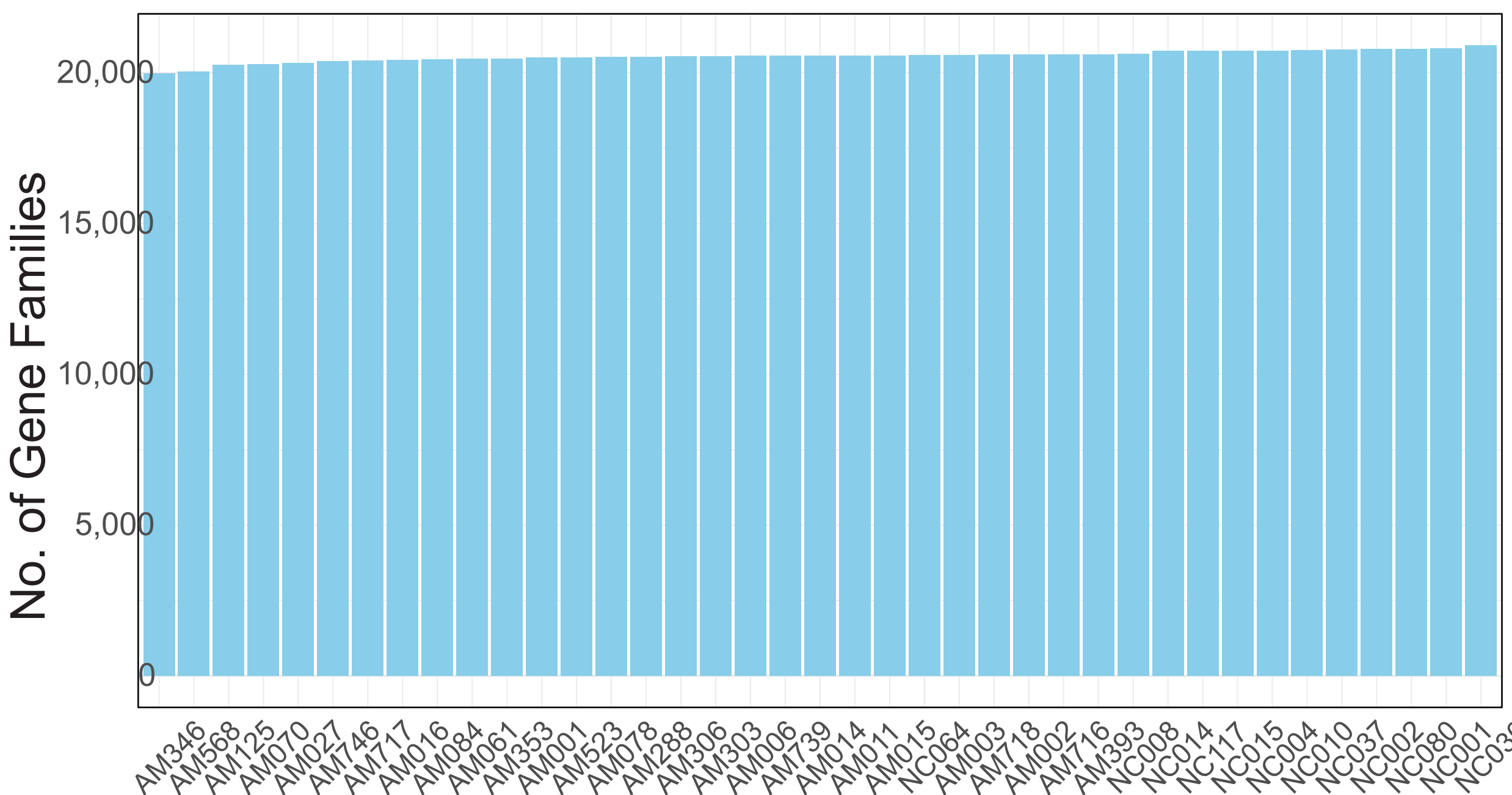
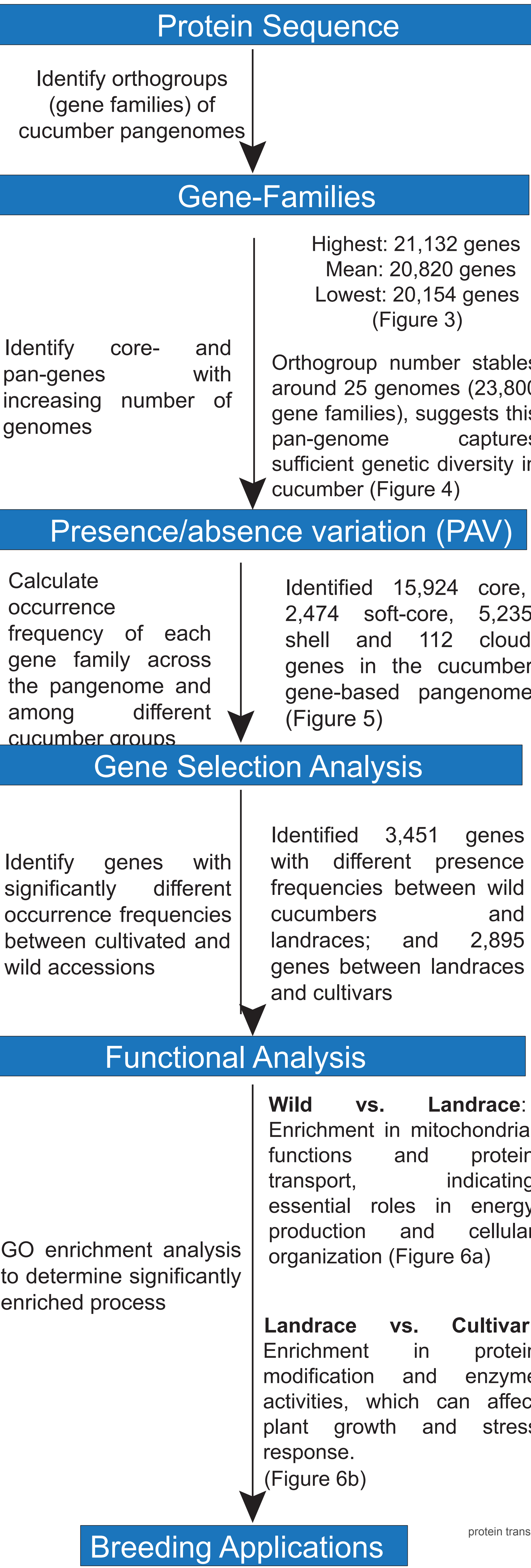


Figure 3. Gene family number in cucumber accessions

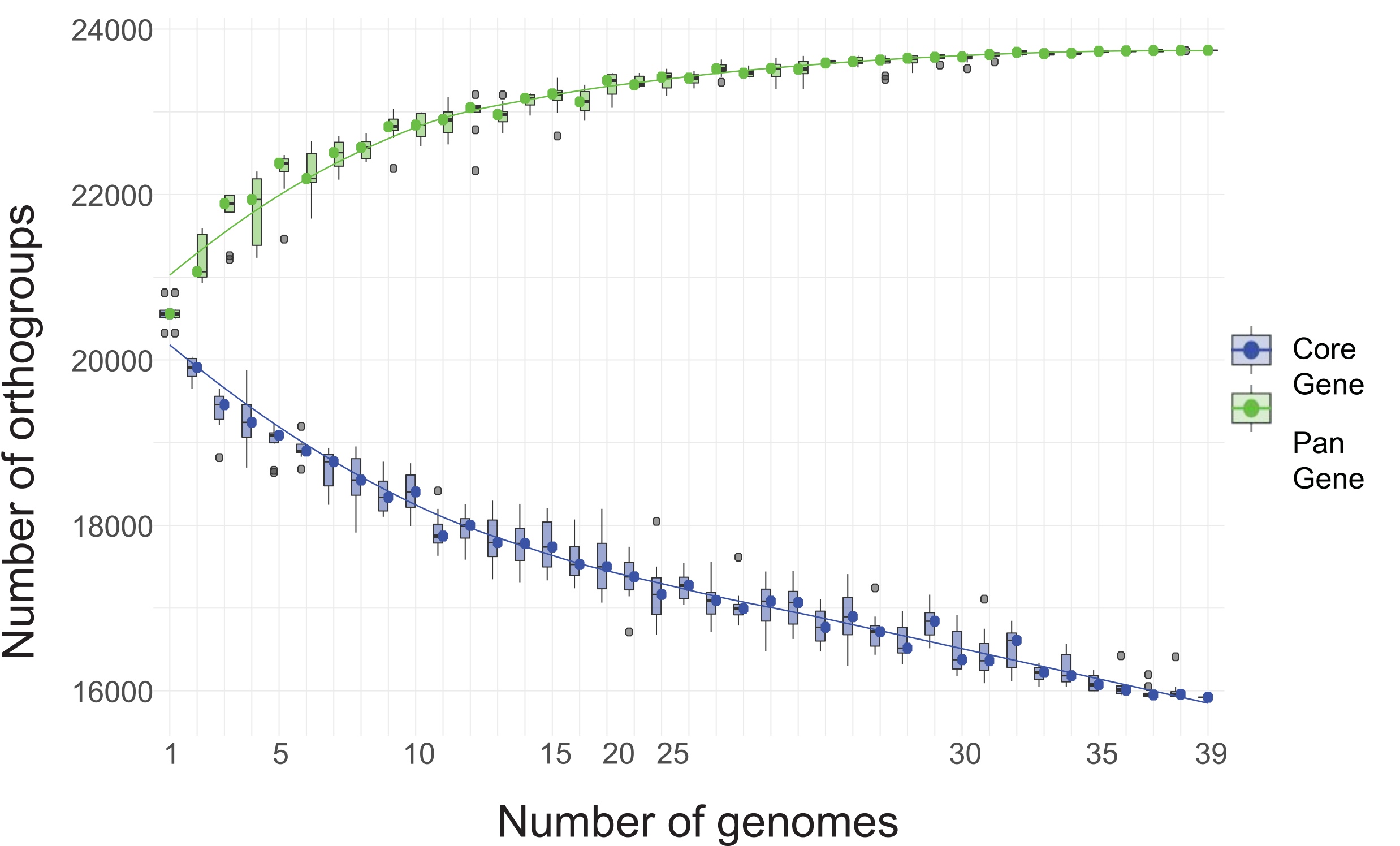


Figure 4. Pan- and core-gene analysis of cucumber pangenome

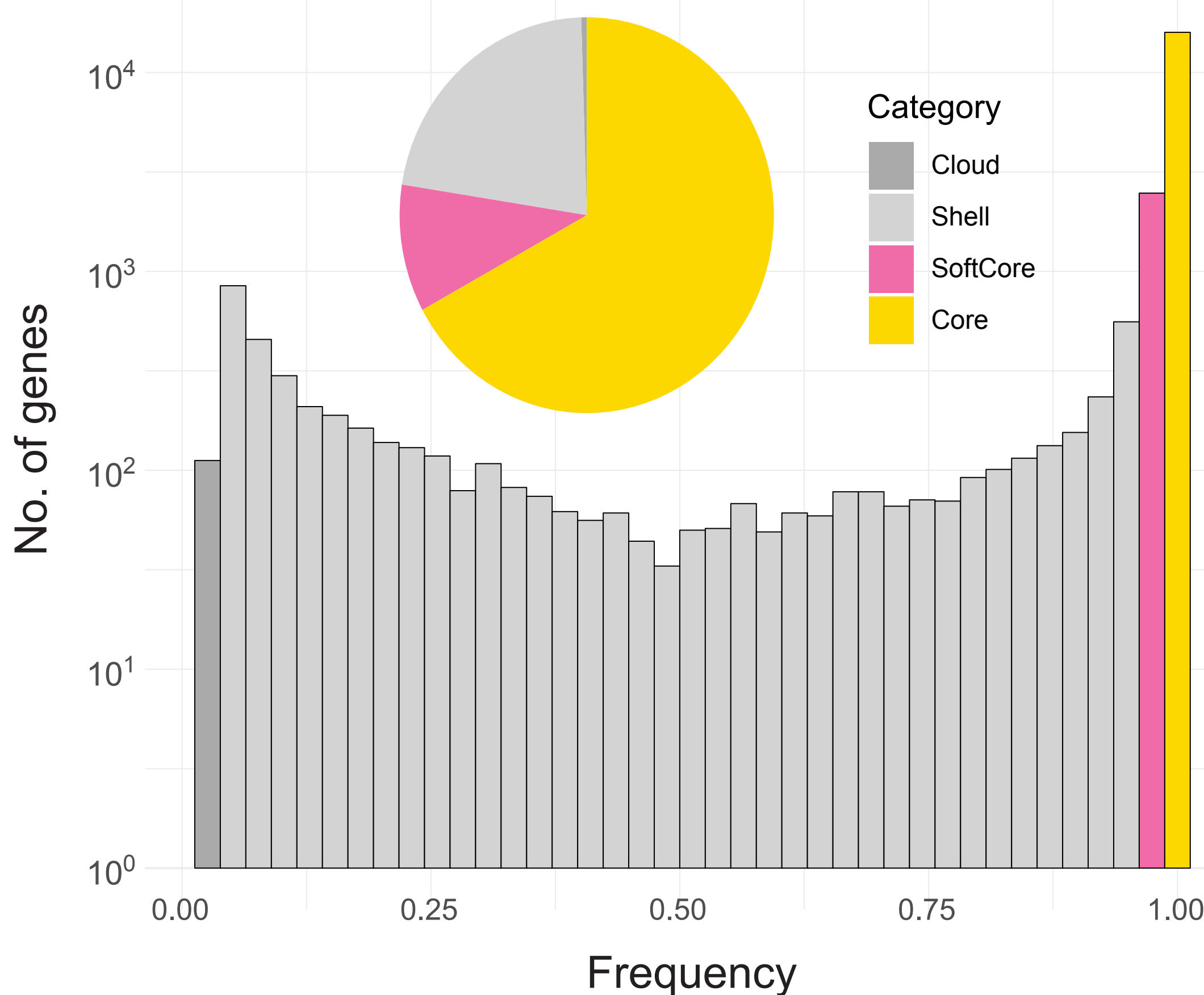
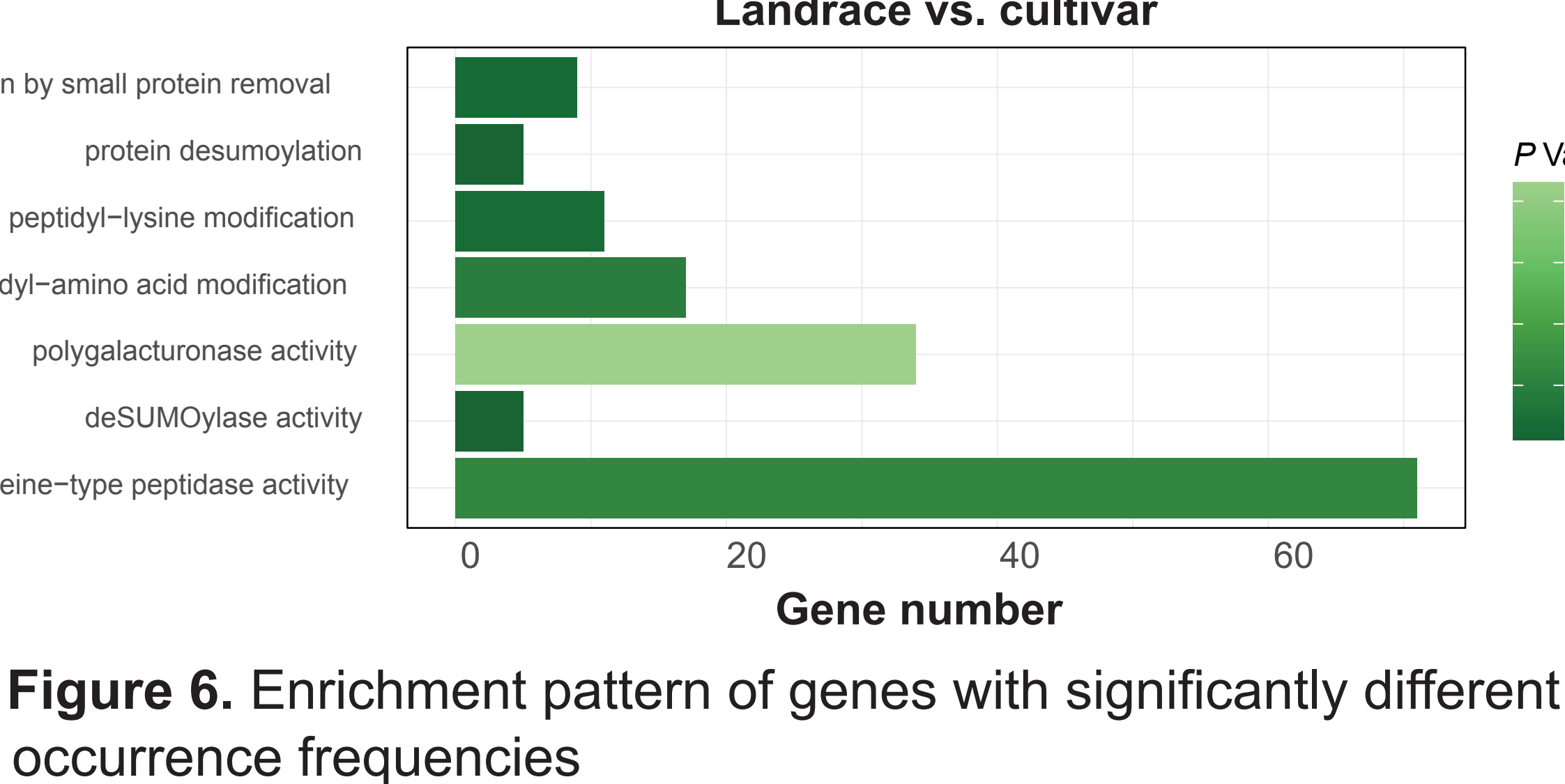
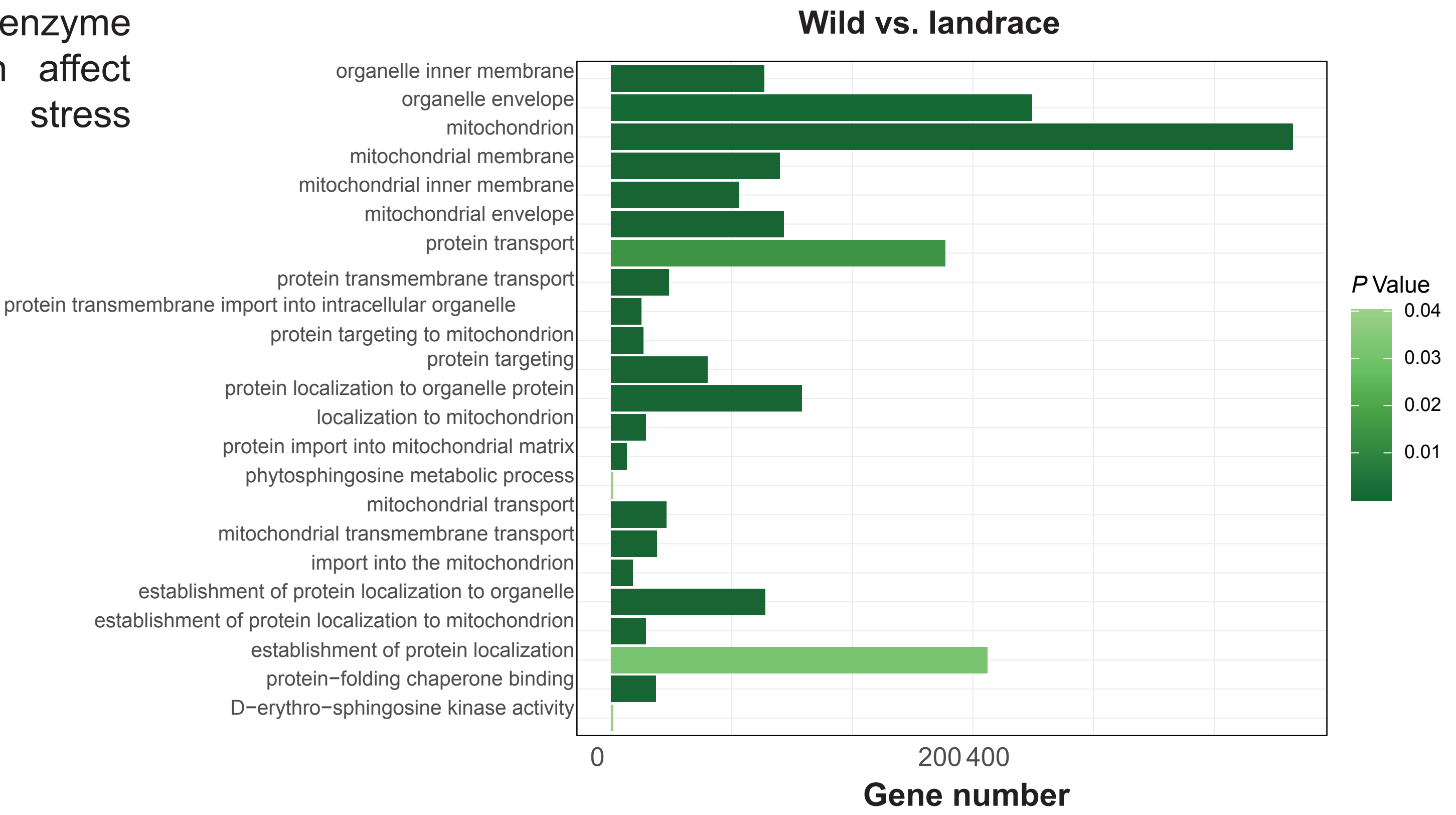


Figure 5. PAV Analysis of cucumber pangenome



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